Day Shift – Prototype Evaluation (CS 435 HCI, Group 1)

Thank you for trying "Day Shift", a webcam-based home rehabilitation game in which you work a day as a house painter: masking tape on the wall marks each arm exercise, your hand is the brush, and paint appears while you follow the tape.

This questionnaire is part of a student HCI course project at IIT Gandhinagar. We are testing the game, not you – there are no right or wrong answers.

• Takes about 10–15 minutes.
• Participation is voluntary; you may skip questions, stop or withdraw at any time without giving a reason.
• Responses are anonymous: you are identified only by a participant code (P1, P2, …). We do not collect your name, email, contact details or any medical/health records.
• The camera image is processed only on the laptop and is never stored or sent. No video or audio is recorded.
• Day Shift is not a medical device. Stop immediately and tell the team member if you feel pain, dizziness or strong tiredness.

* Indicates required question

1. Consent *

   I have read the information above (or it was read to me). I understand that participation is voluntary, that I can stop at any time, and that no sensitive personal or health data will be collected.

   Mark only one oval.

   - ◯ I agree to take part
   - ◯ I do not agree

About you

A few general questions so we can describe who took part. No identifying information is asked.

2. Participant code *

   Given to you by the team member, e.g. P1, P2 … (do not write your name)

   _____

3. Which describes you best? *

Mark only one oval.

- ◯ Older adult (60 years or above)
- ◯ Adult under 60 (trying the game on behalf of older users)
- ◯ Physiotherapist / occupational therapist / clinician
- ◯ Caregiver or family member of an older adult
- ◯ Other: _____

4. Age group *

Mark only one oval.

- ◯ 18–29
- ◯ 30–44
- ◯ 45–59
- ◯ 60–69
- ◯ 70–79
- ◯ 80 or above
- ◯ Prefer not to say

5. Language you used in the game *

Mark only one oval.

- ◯ English
- ◯ Hindi
- ◯ Switched between both

6. How comfortable are you with using computers or smartphones? *

Mark only one oval.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very comfortable

7. How often do you play video or mobile games? *

Mark only one oval.

- ◯ Never
- ◯ Rarely
- ◯ A few times a month
- ◯ Weekly
- ◯ Daily

8. Have you used any exercise or rehabilitation app/game before? *

Mark only one oval.

- ◯ Yes
- ◯ No
- ◯ Not sure

9. How did you play today? *

Mark only one oval.

- ◯ Seated
- ◯ Standing
- ◯ Both

10. Which arm did you exercise with? *

Mark only one oval.

- Right
- Left
- Both (at different times)

Before you start playing

Answer this before the session begins.

11. Right now, how much physical effort or tiredness do you feel? (Borg CR10) *

0 = nothing at all, 3 = moderate, 5 = strong, 7 = very strong, 10 = maximal

Mark only one oval.

	0	1	2	3	4	5	6	7	8	9	10	
Notl	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Maximal

After playing – how easy was each task?

Single Ease Question (SEQ). For each task you did, rate how easy or difficult it was overall.

12. Overall, how difficult or easy was each task? *

1 = very difficult, 7 = very easy

Mark only one oval per row.

	1 – Very difficult	2	3	4	5	6	7 – Very easy	Did not do
T1 – Getting ready (setup check: visible, distance, light)								
T2 – Arm measurement (holding the arm out for 3 seconds)								
T3 – Training / watching the demo stroke								
T4 – Painting one wall (following the tape)								
T5 – Taking a break (pause) and continuing								
T6 – Making the text bigger in Settings								
T7 – Reading and understanding the pay slip								

Overall usability (System Usability Scale)

Please give your first reaction to each statement about the Day Shift game.

13. System Usability Scale *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree
1. I think that I would like to use this game frequently.					
2. I found the game unnecessarily complex.					
3. I thought the game was easy to use.					
4. I think that I would need the support of a technical person to be able to use this game.					
5. I found the various functions in this game were well integrated.					
6. I thought there was too much inconsistency in this game.					
7. I would imagine that most people would learn to use this game very quickly.					
8. I found the					

8. I found the game very cumbersome (awkward) to use.					
9. I felt very confident using the game.					
10. I needed to learn a lot of things before I could get going with this game.					

The painter idea (conceptual model)

How well did the house-painter metaphor help you understand and enjoy the exercises?

14. How much do you agree? *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree
Working as a house painter made the exercises easy to understand.					
The masking tape clearly showed me where to move my arm.					
The direction arrows and the green start circle showed me how to begin a stroke.					
The brush on the screen moved the way my hand moved.					
Paint appearing told me when I was moving correctly.					
The animated demo before each wall					

The wall showed me what to do.					
The clock (9:00 to 17:00) showed me how much work was left.					
Earning a wage made me want to keep going.					
The pay slip clearly separated my pay from my movement summary.					
Exercising felt more purposeful than doing plain repetitions.					

Seeing, hearing and controlling the game

Rate each part of the interface you used. Choose "Did not use / N/A" if you did not try it.

15. Reading and seeing *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree	Did not use / N/A
The text was large and easy to read.	◯	◯	◯	◯	◯	◯
Colours and contrast made everything easy to see.	◯	◯	◯	◯	◯	◯
Buttons were big enough and easy to tell apart.	◯	◯	◯	◯	◯	◯
Each screen had a manageable number of choices.	◯	◯	◯	◯	◯	◯
Icons always had labels that explained them.	◯	◯	◯	◯	◯	◯

16. Controlling the game *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree	Did not use / N/A
Raising my free hand to bring up the pointer was easy.						
Selecting by holding the pointer over a button (dwell ring) was easy.						
The pointer did not select things by accident.						
I could reach all buttons without stretching across my body.						
Voice commands (pause, resume, repeat, next, back, help) worked well.						
Mouse / keyboard / touch worked when I needed them.						

17. Feedback and guidance *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree	Did not use / N/A
Spoken instructions were clear and at a comfortable speed.						
Captions helped me follow what was said.						
Sounds (chimes and tones) were pleasant and easy to hear.						
I always knew what the game was doing and what to do next.						
When a stroke restarted, I understood why.						
The setup check helped me sit at the right place and distance.						
The game kept track of me reliably (few unwanted						

pauses).

18. Which way of controlling the game would you prefer? *

Mark only one oval.

- Free-hand pointer with dwell
- Voice commands
- Mouse or touch screen
- Keyboard
- No preference
- Other:

19. How easy was it to find and change a setting (e.g. text size, theme, language)?

Mark only one oval.

	1	2	3	4	5	
Very						Very easy

How the game made you feel

Day Shift is designed to be respectful of older adults.

20. How much do you agree? *

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree
The game treated me like an adult.					
The wording and visuals were respectful, never childish.					
I did not feel judged or criticised when a stroke had to be redone.					
I would not feel embarrassed using this game in front of family or friends.					
I enjoyed playing the game.					
I felt in control of the game at all times.					
I was not worried about my privacy while the camera was on.					

21. Which word best describes your overall experience? *

Mark only one oval.

- Enjoyable
- Motivating
- Relaxing
- Neutral
- Boring
- Confusing
- Frustrating
- Tiring
- Other: _____

Comfort and effort after playing

22. Right now, how much physical effort or tiredness do you feel? (Borg CR10, after the session) *

0 = nothing at all, 3 = moderate, 5 = strong, 7 = very strong, 10 = maximal

Mark only one oval.

	0	1	2	3	4	5	6	7	8	9	10	
Notl	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Maximal

23. Did you feel any of the following during the session?

Tick all that apply

Check all that apply.

- [ ] None
- [ ] Arm or shoulder tiredness
- [ ] Mild discomfort
- [ ] Pain
- [ ] Dizziness
- [ ] Eye strain
- [ ] Felt rushed or under time pressure
- [ ] Other: _____

24. The pace of the session was … *

Mark only one oval.

- ◯ Too slow
- ◯ About right
- ◯ Too fast

25. The difficulty of the exercises (tape width, accuracy needed) was … *

Mark only one oval.

- ◯ Too easy
- ◯ About right
- ◯ Too hard

26. The length of the shift was … *

Mark only one oval.

- ◯ Too short
- ◯ About right
- ◯ Too long

Overall opinion

27. How likely would you be to use Day Shift at home for your exercises (or recommend it to someone who needs them)? *

Mark only one oval.

	0	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Extremely likely

28. Which features did you find most useful?

Tick up to three

Check all that apply.

- ☐ Painter / wage idea
- ☐ Paint feedback on the tape
- ☐ Spoken instructions and captions
- ☐ Large text and high contrast
- ☐ Free-hand pointer with dwell
- ☐ Voice commands
- ☐ Hindi language option
- ☐ Pause and rest breaks
- ☐ Pay slip and progress summary
- ☐ Automatic arm measurement
- ☐ Other: _____

29. Which trade would you most like to play next?

Mark only one oval.

- Gardener
- Postal sorter
- Carpenter
- Painter is enough for me
- Other: _____

30. What did you like most about the game?

31. What was confusing, difficult or frustrating?

32. If you could change one thing, what would it be?

33. Any other comments?

For physiotherapists / clinicians only (others skip this page)

Questions about the exercises and Expert mode (turning a phone video into an exercise).

34. How much do you agree?

Mark only one oval per row.

	1 – Strongly disagree	2 – Disagree	3 – Neutral	4 – Agree	5 – Strongly agree	Did not use / N/A
The built-in exercises are appropriate for older adults in home rehabilitation.						
The movement paths match standard exercise descriptions.						
Expert mode turned my video into a correct exercise path.						
The angle chart and repetition view helped me check the recording.						
Adjusting tape width and number of strokes gave me enough control.						
The movement summary (strokes, neatness, range of motion) is useful to me.						

The error rules (trunk lean, shoulder hike, leaving the tape) are clinically sensible.						

35. Which exercises or safety checks should be added or changed?

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